

Pemrograman Web Lanjut

Laporan aplikasi SINERGI

Dosen Pengampu :

Asep Taufik Muharram, S.Kom., M.Kom.



Disusun Oleh :

Kelompok 1

Arif Fadillah Wicaksono (2407411007)

Haikal Benyamin Prabowo (2407411025)

Meiza Zafira Angraini (2407411023)

Raditya Meyka Harry Sandhiva (2407411014)

TI 3 A

**PROGRAM STUDI TEKNIK INFORMATIKA
JURUSAN TEKNIK INFORMATIKA DAN KOMPUTER
POLITEKNIK NEGERI JAKARTA**

2025/2026

CRUD Forum

1. Create Forum

The screenshot shows a web interface for creating a new forum. At the top, there's a header with the word "SINERGI" in a light blue box. Below the header, there's a blue square button with the letters "NF" and a "New Forum" button with a public icon. The "New Forum" button is highlighted with a blue border. Below the "New Forum" button, there's a "Forum Details" section with a close button (X). The "Forum Name" field contains the text "Example: Indonesian Coding Community" and a character count "0/30". The "Privacy" section has two options: "Public" (Everyone can see) and "Private" (Members only). The "Public" option is selected. Below the "Privacy" section, there are two upload areas: "Profile Photo" and "Banner / Cover". Both areas have a dashed border and a central icon (a person for the profile photo and a picture for the banner) with the text "Upload Profile" and "Upload Banner" respectively.

```
public function createForum()
{
    header('Content-Type: application/json');
    // set response menjadi JSON

    if (!isset($_SESSION['user_id'])) {
        // jika user belum login
        echo json_encode(['status' => 'error', 'message' => 'Unauthorized: Please login first.']);
        return;
    }

    // ambil input dari form
    $name = isset($_POST['NAME']) ? trim($_POST['NAME']) : "";
    $about = isset($_POST['ABOUT']) ? trim($_POST['ABOUT']) : "";
    $isPrivate = isset($_POST['IS_PRIVATE']) ? intval($_POST['IS_PRIVATE']) : 0;
    $accessKey = isset($_POST['ACCESS_KEY']) ? trim($_POST['ACCESS_KEY']) :
    null;
    $ownerId = $_SESSION['user_id'];

    if (empty($name)) {
```

```

// jika nama forum kosong
echo json_encode(['status' => 'error', 'message' => 'Forum name must be filled.']);
return;
}

if ($isPrivate == 1 && empty($accessKey)) {
// jika forum private tapi akses key tidak diisi
echo json_encode(['status' => 'error', 'message' => 'Private forums require an access
key.']);
return;
}

// default value foto & thumbnail
$pathPhoto = null;
$pathThumbnail = null;

// folder penyimpanan
$uploadDir = __DIR__ . '/../storage/forums/photos/';
$uploadDirThumb = __DIR__ . '/../storage/forums/thumbnail/';

// buat folder jika belum ada
if (!is_dir($uploadDir)) mkdir($uploadDir, 0777, true);
if (!is_dir($uploadDirThumb)) mkdir($uploadDirThumb, 0777, true);

// upload foto profile forum
if (isset($_FILES['PATH_PHOTO']) && $_FILES['PATH_PHOTO']['error'] ===
UPLOAD_ERR_OK) {
    $pathPhoto = $this->uploadFile($_FILES['PATH_PHOTO'], $uploadDir, 'profile');
    if (!$pathPhoto) {
        echo json_encode(['status' => 'error', 'message' => 'Failed to upload profile photo
(incorrect format/size).']);
        return;
    }
}

// upload thumbnail/banner forum
if (isset($_FILES['PATH_THUMBNAIL']) &&
$_FILES['PATH_THUMBNAIL']['error'] === UPLOAD_ERR_OK) {
    $pathThumbnail = $this->uploadFile($_FILES['PATH_THUMBNAIL'],
$uploadDirThumb, 'banner');
    if (!$pathThumbnail) {
        echo json_encode(['status' => 'error', 'message' => 'Failed to upload banner
(incorrect format/size).']);
        return;
    }
}

// role menentukan status forum (langsung aktif atau pending)
$role = $_SESSION['role'] ?? null;

```

```
$statusForums = ($role === "ADMIN" || $role === "DOSEN") ? "ACTIVE" :
"NONACTIVE";
```

```
// data yang akan disimpan
```

```
$data = [
    'NAME' => $name,
    'ABOUT' => $about,
    'IS_PRIVATE' => $isPrivate,
    'ACCESS_KEY' => ($isPrivate == 1) ? $accessKey : null,
    'OWNER_ID' => $ownerId,
    'PATH_PHOTO' => $pathPhoto,
    'PATH_THUMBNAIL' => $pathThumbnail,
    'STATUS' => $statusForums
];
```

```
try {
```

```
    // simpan ke database via model
```

```
    $result = $this->forumModel->createForum($data);
```

```
    if ($result['success']) {
```

```
        // sukses membuat forum
```

```
        echo json_encode([
            'status' => 'success',
            'message' => 'Forum successfully created!',
            'id' => $result['ID']
        ]);
```

```
    } else {
```

```
        // gagal simpan database
```

```
        echo json_encode([
            'status' => 'error',
            'message' => 'Failed to save to the database.',
            'error' => $result['error']
        ]);
```

```
    }
```

```
} catch (Exception $e) {
```

```
    // jika terjadi error internal
```

```
    echo json_encode(['status' => 'error', 'message' => 'Server Error: ' .
    $e->getMessage()]);
}
```

```
}
```

Modelnya

```
public function createForum($data)
```

```
{
```

```
    $conn = self::getConnection();
```

```
    // koneksi database Oracle
```

```

$forumId = uniqid();
// generate ID unik untuk forum

$ownerId = $_SESSION['user_id'];
// ambil ID user yang buat forum

// query insert forum
$sql = "INSERT INTO FORUMS (
    ID, NAME, PATH_THUMBNAIL, PATH_PHOTO, ABOUT, IS_PRIVATE,
    ACCESS_KEY, OWNER_ID, STATUS, CREATED_AT
) VALUES (
    :id, :name, :path_thumbnail, :path_photo, :about,
    :is_private, :access_key, :owner_id, :status_forum, SYSDATE
)";

$stmt = oci_parse($conn, $sql); // prepare query

// bind parameter
oci_bind_by_name($stmt, ":id", $forumId);
oci_bind_by_name($stmt, ":name", $data['NAME']);
oci_bind_by_name($stmt, ":path_thumbnail", $data['PATH_THUMBNAIL']);
oci_bind_by_name($stmt, ":path_photo", $data['PATH_PHOTO']);
oci_bind_by_name($stmt, ":about", $data['ABOUT']);
oci_bind_by_name($stmt, ":is_private", $data['IS_PRIVATE']);
oci_bind_by_name($stmt, ":access_key", $data['ACCESS_KEY']);
oci_bind_by_name($stmt, ":owner_id", $ownerId);
oci_bind_by_name($stmt, ":status_forum", $data['STATUS']);

$result = oci_execute($stmt, OCI_NO_AUTO_COMMIT);
// jalankan query tanpa commit otomatis

if (!$result) {
    // jika gagal insert forum
    $e = oci_error($stmt);
    oci_rollback($conn); // batalkan transaksi
    oci_free_statement($stmt);

    return [
        'success' => false,
        'error' => $e['message']
    ];
}

oci_free_statement($stmt);

// setelah forum dibuat, tambahkan owner sebagai member

$memberId = uniqid(); // ID unik member
$status = "JOINED"; // status member owner otomatis JOINED

```

```

$sql2 = "INSERT INTO FORUM_MEMBERS (ID, FORUM_ID, USER_ID, STATUS,
JOINED_AT)
VALUES (:id, :forum_id, :user_id, :status ,SYSDATE)";

$stmt2 = oci_parse($conn, $sql2);

// bind parameter insert member
oci_bind_by_name($stmt2, ":id", $memberId);
oci_bind_by_name($stmt2, ":forum_id", $forumId);
oci_bind_by_name($stmt2, ":status", $status);
oci_bind_by_name($stmt2, ":user_id", $ownerId);

$result2 = oci_execute($stmt2, OCI_NO_AUTO_COMMIT);

if (!$result2) {
    // jika gagal insert ke FORUM_MEMBERS
    $e = oci_error($stmt2);
    oci_rollback($conn);
    oci_free_statement($stmt2);

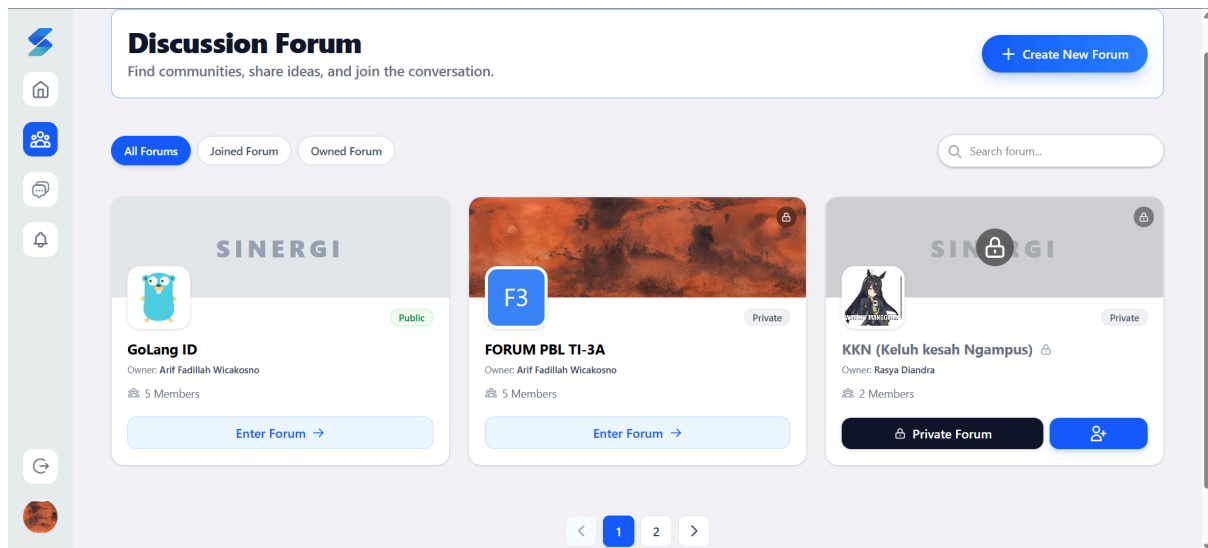
    return [
        'success' => false,
        'error' => $e['message']
    ];
}

// jika semua berhasil → commit transaksi
oci_commit($conn);
oci_free_statement($stmt2);

return [
    'success' => true,
    'ID' => $forumId
];
}

```

2. Read Forum



```
public function index()
{
    $myUserId = $_SESSION['user_id'] ?? null;
    // ambil user_id jika login

    $userRole = $_SESSION['role'] ?? '';
    // ambil role user

    if (in_array($userRole, ['MITRA', 'ALUMNI'])) {
        // kalau role MITRA atau ALUMNI, filter langsung dipaksa jadi "joined"
        $filter = 'joined';
        $search = ''; // search dikosongkan (tidak boleh searching)
    } else {
        // selain itu boleh pakai filter & search dari URL
        $filter = $_GET['filter'] ?? 'all';
        $search = $_GET['search'] ?? '';
    }

    // pagination
    $page = isset($_GET['page']) ? (int)$_GET['page'] : 1;
    $limit = 6; // jumlah item per halaman

    if ($page < 1) $page = 1; // minimal page 1

    $offset = ($page - 1) * $limit;
    // hitung offset untuk query database

    // ambil data forum sesuai filter + search + pagination
    $forums = $this->forumModel->getForumsWithFilter($myUserId, $filter, $search,
```

```

$limit, $offset);

    // hitung total forum untuk pagination
    $totalForums = $this->forumModel->countForumsWithFilter($myUserId, $filter,
$search);
    $totalPages = ceil($totalForums / $limit);

    // load halaman view utama forum
    $contentViewForum = __DIR__ . '/../views/forum/index.php';
    require_once __DIR__ . '/../views/forum/layout.php';
}

```

Modelnya

```

public function getForumsWithFilter($userId, $filter, $search, $limit, $offset)
{
    $conn = self::getConnection();
    // mulai koneksi ke DB

    $bindings = [];
    // tempat simpan semua parameter bind

    // query utama select forum
    $sql = "SELECT
        f.*,
        u.FULL_NAME AS OWNER_NAME,
        // hitung jumlah member yg joined pada forum itu
        (SELECT COUNT(*) FROM FORUM_MEMBERS fm WHERE fm.FORUM_ID =
f.ID AND fm.STATUS = 'JOINED') AS TOTAL_MEMBERS,
        // cek apakah user saat ini sudah join forum (1) atau belum (0)
        (SELECT COUNT(*) FROM FORUM_MEMBERS fm2 WHERE fm2.FORUM_ID
= f.ID AND fm2.USER_ID = :userid_check AND fm2.STATUS = 'JOINED') AS
IS_MEMBER
    FROM FORUMS f
    JOIN USERS u ON f.OWNER_ID = u.ID ";

    // jika filter = joined, tambahkan join ekstra ke forum_members
    if ($filter === 'joined') {
        $sql .= " JOIN FORUM_MEMBERS fm_filter ON f.ID = fm_filter.FORUM_ID ";
    }

    // filter status forum
    if ($filter === 'owned') {
        // owner bisa lihat forum ACTIVE dan NONACTIVE
        $sql .= " WHERE f.STATUS IN ('ACTIVE', 'NONACTIVE') ";
    } else {
        // user biasa hanya lihat ACTIVE
        $sql .= " WHERE f.STATUS = 'ACTIVE' ";
    }
}

```



```

// filter untuk forum yang user sudah join
if ($filter === 'joined') {
    $sql .= " AND fm_filter.USER_ID = :userid_filter ";
    $sql .= " AND fm_filter.STATUS = 'JOINED' ";
    $bindings[':userid_filter'] = $userId;
}
// filter untuk forum yang dimiliki user
elseif ($filter === 'owned') {
    $sql .= " AND f.OWNER_ID = :owner_id ";
    $bindings[':owner_id'] = $userId;
}

// filter search (mencari nama / description)
if (!empty($search)) {
    $sql .= " AND (LOWER(f.NAME) LIKE '%' || LOWER(:search_query) || '%'"
        . " OR LOWER(f.ABOUT) LIKE '%' || LOWER(:search_desc) || '%') ";
    $bindings[':search_query'] = $search;
    $bindings[':search_desc'] = $search;
}

// urutkan forum yang sudah dijoin user → muncul di atas
$sql .= " ORDER BY IS_MEMBER DESC, f.CREATED_AT DESC"
    . " OFFSET :offset_val ROWS FETCH NEXT :limit_val ROWS ONLY";

$bindings[':offset_val'] = $offset;
$bindings[':limit_val'] = $limit;
$bindings[':userid_check'] = $userId;

$stmt = oci_parse($conn, $sql);

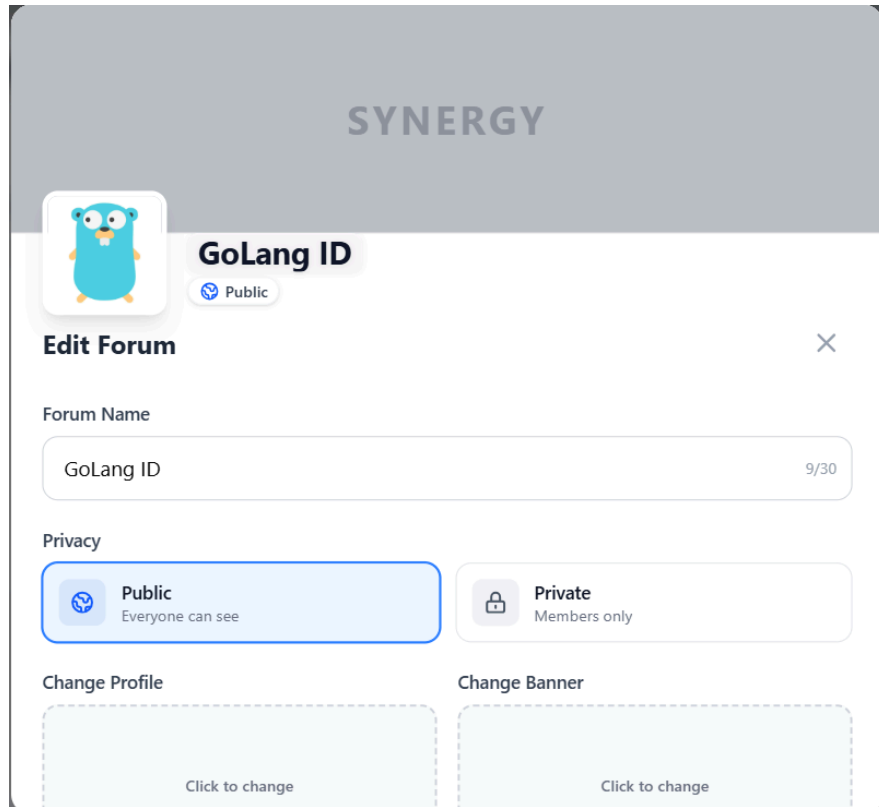
// bind semua parameter
foreach ($bindings as $key => $val) {
    oci_bind_by_name($stmt, $key, $bindings[$key]); // binding dengan reference array
}

oci_execute($stmt);

// ambil semua hasil query
$forums = [];
while ($row = oci_fetch_array($stmt, OCI_ASSOC + OCI_RETURN_NULLS)) {
    $forums[] = $row;
}
return $forums;
}

```

3. Update Forum



```
public function update()
{
    header('Content-Type: application/json'); // Set response menjadi JSON

    // Cek apakah user sudah login
    if (!isset($_SESSION['user_id'])) {
        echo json_encode(['success' => false, 'message' => 'Unauthorized']);
        exit;
    }

    // Ambil ID forum dari form
    $forumId = $_POST['ID'] ?? null;
    if (!$forumId) {
        echo json_encode(['success' => false, 'message' => 'ID Forum not found.']);
        exit;
    }

    // Ambil data forum lama dari database
    $oldForumData = $this->forumModel->getForumById($forumId);
    if (!$oldForumData) {
        echo json_encode(['success' => false, 'message' => 'Forum not found.']);
        exit;
    }

    // Cek apakah user adalah owner forum
```

```

if ($oldForumData['OWNER_ID'] !== $_SESSION['user_id']) {
    echo json_encode(['success' => false, 'message' => 'You are not the owner of this
forum!']);
    exit;
}

// Ambil data private dari form (0/1)
$isPrivate = $_POST['IS_PRIVATE'] ?? 0;

// Data yang akan diupdate
$updateData = [
    'NAME'      => $_POST['NAME'],
    'ABOUT'    => $_POST['ABOUT'],
    'IS_PRIVATE' => $isPrivate,
    'ACCESS_KEY' => ($isPrivate == 1) ? $_POST['ACCESS_KEY'] : null
];

// =====
// UPLOAD FOTO PROFILE
// =====
if (isset($_FILES['PATH_PHOTO']) && $_FILES['PATH_PHOTO']['error'] ===
UPLOAD_ERR_OK) {
    $file = $_FILES['PATH_PHOTO'];

    // Ambil ekstensi file
    $ext = pathinfo($file['name'], PATHINFO_EXTENSION);

    // Bikin nama file baru unik
    $newFileName = 'forum_profile_' . uniqid() . '.' . $ext;

    // Lokasi penyimpanan
    $uploadDir = __DIR__ . '/../storage/forums/photos/';

    // Pindahkan file ke folder storage
    if (move_uploaded_file($file['tmp_name'], $uploadDir . $newFileName)) {

        // Masukkan nama foto baru ke database
        $updateData['PATH_PHOTO'] = $newFileName;

        // Hapus foto lama kalau ada
        if (!empty($oldForumData['PATH_PHOTO'])) {
            $oldFile = $uploadDir . $oldForumData['PATH_PHOTO'];
            if (file_exists($oldFile)) unlink($oldFile);
        }
    }
}

// =====
// UPLOAD FOTO THUMBNAIL

```

```

// =====
if (isset($_FILES['PATH_THUMBNAIL']) &&
$_FILES['PATH_THUMBNAIL']['error'] === UPLOAD_ERR_OK) {
    $file = $_FILES['PATH_THUMBNAIL'];
    $ext = pathinfo($file['name'], PATHINFO_EXTENSION);

    $newFileName = 'forum_banner_' . uniqid() . '.' . $ext;
    $uploadDir = __DIR__ . '/../storage/forums/thumbnaill/';

    if (move_uploaded_file($file['tmp_name'], $uploadDir . $newFileName)) {
        $updateData['PATH_THUMBNAIL'] = $newFileName;

        // Hapus thumbnail lama
        if (!empty($oldForumData['PATH_THUMBNAIL'])) {
            $oldFile = $uploadDir . $oldForumData['PATH_THUMBNAIL'];
            if (file_exists($oldFile)) unlink($oldFile);
        }
    }
}

// Jalankan update ke database
if ($this->forumModel->updateForum($forumId, $updateData)) {
    echo json_encode(['success' => true, 'message' => 'Changes saved successfully!']);
} else {
    echo json_encode(['success' => false, 'message' => 'Failed saving to database.']);
}
}

```

Modelnya

```

public function updateForum($id, $data)
{
    $conn = self::getConnection();

    // Query dasar update
    $sql = "UPDATE FORUMS SET
        NAME = :name,
        ABOUT = :about,
        IS_PRIVATE = :is_private,
        ACCESS_KEY = :access_key";

    // Jika ada foto profile baru, tambahkan ke query
    if (array_key_exists('PATH_PHOTO', $data)) {
        $sql .= ", PATH_PHOTO = :photo";
    }

    // Jika ada thumbnail baru
    if (array_key_exists('PATH_THUMBNAIL', $data)) {
        $sql .= ", PATH_THUMBNAIL = :thumb";
    }
}

```

```

}

// WHERE untuk pilih forum by ID
$sql .= " WHERE ID = :id";

$stmt = oci_parse($conn, $sql);

// Binding data utama
oci_bind_by_name($stmt, ":id", $id);
oci_bind_by_name($stmt, ":name", $data['NAME']);
oci_bind_by_name($stmt, ":about", $data['ABOUT']);
oci_bind_by_name($stmt, ":is_private", $data['IS_PRIVATE']);
oci_bind_by_name($stmt, ":access_key", $data['ACCESS_KEY']);

// Binding image kalau ada
if (array_key_exists('PATH_PHOTO', $data)) {
    oci_bind_by_name($stmt, ":photo", $data['PATH_PHOTO']);
}

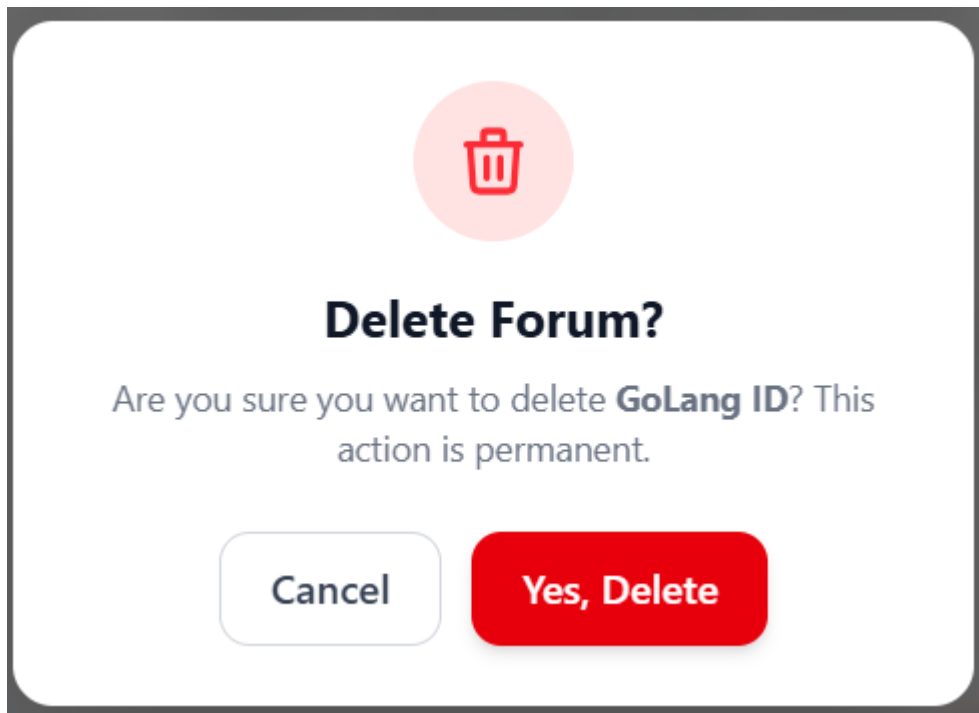
if (array_key_exists('PATH_THUMBNAIL', $data)) {
    oci_bind_by_name($stmt, ":thumb", $data['PATH_THUMBNAIL']);
}

// Eksekusi query tanpa auto commit
$result = oci_execute($stmt, OCI_NO_AUTO_COMMIT);

// Kalau sukses → commit
if ($result) {
    oci_commit($conn);
    return true;
} else {
    return false; // gagal update
}
}

```

4. Delete Forum



```
public function delete()
{
    header('Content-Type: application/json'); // Response harus JSON

    // =====
    // CEK LOGIN USER
    // =====
    if (!isset($_SESSION['user_id'])) {
        echo json_encode(['success' => false, 'message' => 'Unauthorized']);
        exit;
    }

    // Ambil ID forum dari POST
    $forumId = $_POST['ID'] ?? null;
    if (!$forumId) {
        echo json_encode(['success' => false, 'message' => 'ID Forum not found.']);
        exit;
    }

    // Ambil data forum untuk pengecekan
    $forum = $this->forumModel->getForumById($forumId);

    // Forum tidak ditemukan
    if (!$forum) {
        echo json_encode(['success' => false, 'message' => 'Forum not found']);
        exit;
    }
}
```

```

// Pastikan user adalah OWNER forum
if ($forum['OWNER_ID'] !== $_SESSION['user_id']) {
    echo json_encode(['success' => false, 'message' => 'You are not the owner of this
forum!']);
    exit;
}

// Ambil semua file yang terkait forum (thumb, photo, topic media)
$assets = $this->forumModel->getAllForumAssets($forumId);

// =====
// MULAI HAPUS DATA DI DATABASE
// =====
if ($this->forumModel->deleteForum($forumId)) {

    // =====
    // HAPUS FILE THUMBNAIL
    // =====
    if (!empty($assets['forum_thumb'])) {
        $path = __DIR__ . '/../storage/forums/thumbnail/' . $assets['forum_thumb'];
        if (file_exists($path)) unlink($path); // hapus file fisik
    }

    // =====
    // HAPUS FOTO PROFIL FORUM
    // =====
    if (!empty($assets['forum_photo'])) {
        $path = __DIR__ . '/../storage/forums/photos/' . $assets['forum_photo'];
        if (file_exists($path)) unlink($path);
    }

    // =====
    // HAPUS SEMUA MEDIA TOPIC
    // =====
    if (!empty($assets['topic_media'])) {
        foreach ($assets['topic_media'] as $mediaFile) {
            $path = __DIR__ . '/../storage/topics/media/' . $mediaFile;
            if (file_exists($path)) unlink($path);
        }
    }

    // Berhasil
    echo json_encode(['success' => true, 'message' => 'The forum and all its contents have
been successfully deleted.']);

} else {
    // Gagal hapus data di database
    echo json_encode(['success' => false, 'message' => 'Failed to delete data from the
database.']);
}

```

```
}  
}
```

Modelnya

```
public function deleteForum($id)  
{  
    $conn = self::getConnection();  
  
    try {  
        // =====  
        // HAPUS REPORT YANG MENYERANG FORUM  
        // =====  
        $sqlReport = "DELETE FROM REPORT WHERE TARGET_ID = :id AND  
TARGET_TYPE = 'FORUM'";  
        $stmtReport = oci_parse($conn, $sqlReport);  
        oci_bind_by_name($stmtReport, ":id", $id);  
  
        // Jika gagal hapus report → throw error  
        if (!oci_execute($stmtReport, OCI_NO_AUTO_COMMIT))  
            throw new Exception("Gagal menghapus report forum");  
  
        // Bebaskan statement  
        oci_free_statement($stmtReport);  
  
        // =====  
        // HAPUS SEMUA MEMBER FORUM  
        // =====  
        $sqlMember = "DELETE FROM FORUM_MEMBERS WHERE FORUM_ID = :id";  
        $stmtMember = oci_parse($conn, $sqlMember);  
        oci_bind_by_name($stmtMember, ":id", $id);  
  
        if (!oci_execute($stmtMember, OCI_NO_AUTO_COMMIT))  
            throw new Exception("Gagal menghapus member");  
  
        // =====  
        // HAPUS FORUMNYA  
        // =====  
        $sqlForum = "DELETE FROM FORUMS WHERE ID = :id";  
        $stmtForum = oci_parse($conn, $sqlForum);  
        oci_bind_by_name($stmtForum, ":id", $id);  
  
        if (!oci_execute($stmtForum, OCI_NO_AUTO_COMMIT))  
            throw new Exception("Gagal menghapus forum");  
  
        // =====  
        // SEMUA SUKSES → COMMIT  
        // =====  
        oci_commit($conn);  
    }  
}
```



```

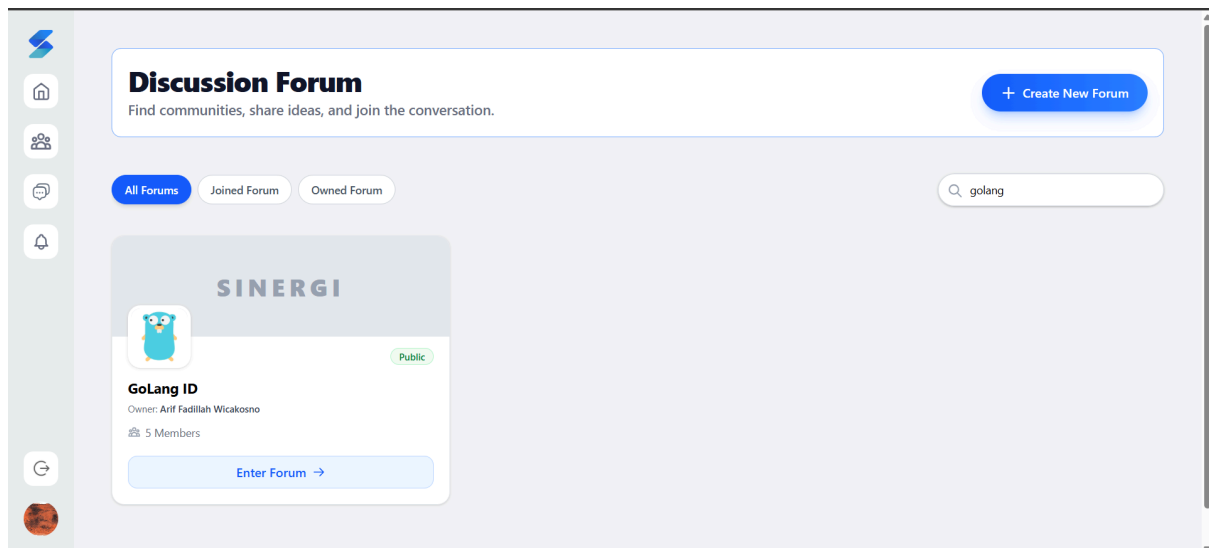
return true;

} catch (Exception $e) {

    // Jika salah satu DELETE gagal → rollback semua perubahan
    oci_rollback($conn);
    return false;
}
}
}

```

Search



HTML-Nya

```

<form action="" method="GET" class="w-full md:w-auto relative bg-white rounded-full shadow-sm">

```

```

    <!-- input hidden untuk membawa value filter di URL -->

```

```

    <input type="hidden" name="filter" value="<?= htmlspecialchars($filter) ?>">

```

```

    <div class="relative group">

```

```

        <!-- input search -->

```

```

        <input type="text"

```

```

            name="search"

```

```

            value="<?= htmlspecialchars($search) ?>"

```

```

            placeholder="Search forum..."

```

```

            class="w-full md:w-72 pl-10 pr-4 py-2.5 rounded-full

```

```

                border border-gray-300

```

```

                focus:outline-none focus:ring-2 focus:ring-blue-500

```

```

                focus:border-transparent text-sm transition-all shadow-sm">

```

```

    <!-- icon search di dalam input -->

```

```

    <div class="absolute left-3 top-3 text-gray-400

```

```

        group-focus-within:text-blue-500 transition-colors">
        <svg class="w-5 h-5" fill="none" stroke="currentColor" viewBox="0 0 24 24">
        <path stroke-linecap="round"
        stroke-linejoin="round"
        stroke-width="2"
        d="M21 21l-6-6m2-5a7 7 0 11-14 0 7 7 0 0114 0z">
        </path>
        </svg>
        </div>
        </div>
    </form>

```

Controllernya

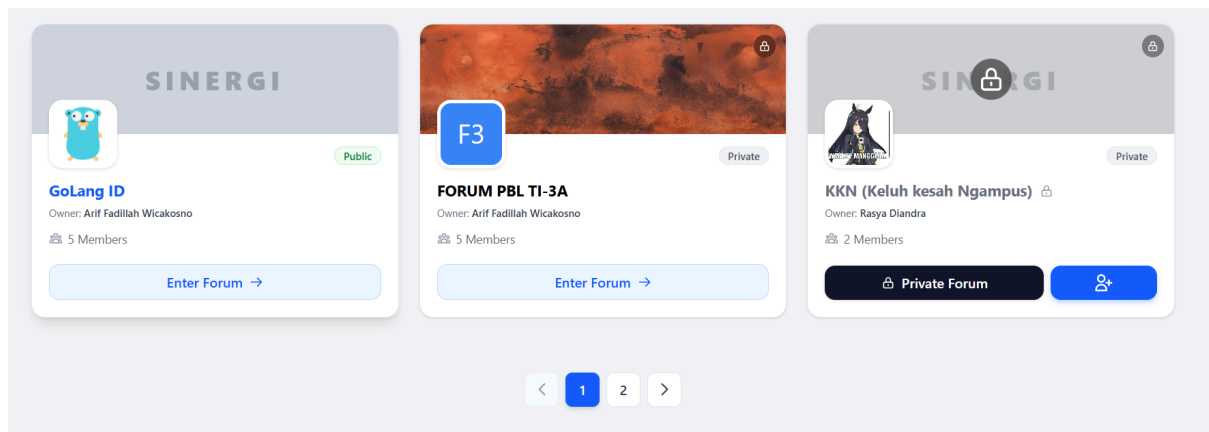
```

if (in_array($userRole, ['MITRA', 'ALUMNI'])) {
    $filter = 'joined'; // jika role MITRA atau ALUMNI, filter dipaksa menjadi "joined"
    $search = ""; // search dikosongkan (mereka tidak bisa melakukan pencarian)
} else {
    $filter = $_GET['filter'] ?? 'all'; // ambil filter dari URL, kalau tidak ada pakai 'all'
    $search = $_GET['search'] ?? ""; // ambil kata pencarian dari URL, kalau tidak ada
    kosong
}

$forums = $this->forumModel->getForumsWithFilter($myUserId, $filter, $search, $limit,
$offset); // Lalu di ambil data forum sesuai keyword dari $search di atas

```

Pagination



Controllernya

```

$page = isset($_GET['page']) ? (int)$_GET['page'] : 1;
// ambil nomor halaman dari URL (?page=2)
// kalau tidak ada, default halaman = 1
// (int) untuk memastikan tipe datanya integer

```

```

$limit = 3;
// jumlah data forum yang ditampilkan per halaman

if ($page < 1) $page = 1;
// jika page kurang dari 1 (misal user ubah URL jadi page=0), dipaksa jadi 1

$offset = ($page - 1) * $limit;
// hitung dari posisi data mana query dimulai
// contoh: page 1 → offset 0
//         page 2 → offset 3
//         page 3 → offset 6

$forums = $this->forumModel->getForumsWithFilter($myUserId, $filter, $search, $limit,
$offset);
// ambil data forum berdasarkan limit & offset (data per halaman)

$totalForums = $this->forumModel->countForumsWithFilter($myUserId, $filter, $search);
// hitung total forum sesuai filter & search (jumlah keseluruhan data)

$totalPages = ceil($totalForums / $limit);
// total halaman = total data dibagi jumlah data per halaman
// ceil() supaya dibulatkan ke atas

```

HTML-Nya

```

<?php if ($totalPages > 1): ?>
    <!-- tampilkan pagination hanya jika total halaman lebih dari 1 -->

    <div class="flex justify-center items-center gap-2 mb-10 pb-10 lg:pb-0">

        <!-- Tombol Previous -->
        <a href="<?= ($page > 1) ? buildUrl('page', $page - 1) : '#' ?>"
            class="w-10 h-10 flex items-center justify-center bg-white border border-gray-200
rounded-lg
            text-gray-700 transition
            <?= ($page > 1) ? 'hover:bg-gray-50 hover:text-blue-600' : 'opacity-50
cursor-not-allowed' ?>">
            <!-- jika page > 1 → link ke halaman sebelumnya
            jika page == 1 → tombol disable (#) -->

            <svg class="w-5 h-5" ...>
                <path d="M15.75 19.5L8.25 12l7.5-7.5" />
                <!-- icon panah ke kiri -->
            </svg>
        </a>

        <!-- Nomor halaman -->
        <?php for ($i = 1; $i <= $totalPages; $i++): ?>

```

```

<a href="<?= buildUrl('page', $i) ?>"
    class="w-10 h-10 flex items-center justify-center rounded-lg border transition
font-medium
    <?= $i == $page
        ? 'bg-blue-600 text-white border-blue-600 shadow-md' // jika nomor ini
adalah halaman saat ini → highlight
        : 'bg-white text-gray-700 border-gray-200 hover:bg-gray-50
hover:text-blue-600' ?>">
    <?= $i ?> <!-- menampilkan nomor halaman -->
</a>
<?php endfor; ?>

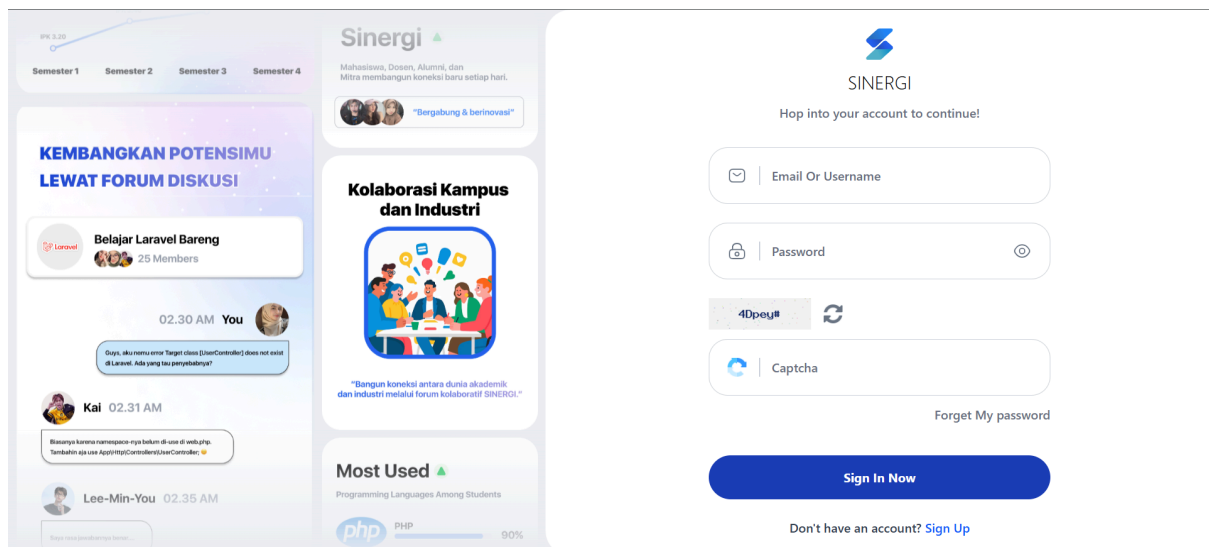
<!-- Tombol Next -->
<a href="<?= ($page < $totalPages) ? buildUrl('page', $page + 1) : '#' ?>"
    class="w-10 h-10 flex items-center justify-center bg-white border border-gray-200
rounded-lg
    text-gray-700 transition
    <?= ($page < $totalPages)
        ? 'hover:bg-gray-50 hover:text-blue-600'
        : 'opacity-50 cursor-not-allowed' ?>">
    <!-- jika masih ada halaman berikutnya → ke page + 1
    jika page terakhir → disable (#) -->

<svg class="w-5 h-5" ...>
    <path d="M8.25 4.517.5 7.5 7.5" />
    <!-- icon panah ke kanan -->
</svg>
</a>

</div>
<?php endif; ?>

```

Captcha



```

<?php
session_start(); // Mulai session untuk mengambil captcha yang sudah disimpan

// Ambil captchanya dari session, kalau tidak ada tampilkan "ERROR"
$captcha = $_SESSION['captcha'] ?? 'ERROR';

// Ukuran gambar captcha
$width = 130;
$height = 40;

// Buat canvas gambar
$image = imagecreate($width, $height);

// =====
// WARNA BACKGROUND dan TEXT
// =====
$bg = imagecolorallocate($image, 240, 240, 240); // warna background
$textColor = imagecolorallocate($image, 20, 40, 100); // warna tulisan captcha

// =====
// TAMBAH NOISE (acak-acak pixel)
// =====
for ($i = 0; $i < 50; $i++) {
    $noiseColor = imagecolorallocate($image, rand(150, 255), rand(150, 255), rand(150, 255));
    imagesetpixel($image, rand(0, $width), rand(0, $height), $noiseColor);
}

// =====
// TULIS CAPTCHA KE GAMBAR
// =====

// ukuran font bawaan (1–5, semakin besar semakin tebal)
$fontSize = 5;

// hitung posisi biar teks berada di tengah gambar
$x = (imagesx($image) - imagefontwidth($fontSize) * strlen($captcha)) / 2;
$y = (imagesy($image) - imagefontheight($fontSize)) / 2;

// gambar teks captcha
imagestring($image, $fontSize, $x, $y, $captcha, $textColor);

// Kirim header gambar PNG
header('Content-Type: image/png');

// output gambar
imagepng($image);

```

```
// hapus dari memory RAM server
imagedestroy($image);
?>
```

HTMLNYa

```
<div>
  <div class="flex gap-3 items-center mb-3">
    <!-- Gambar captcha, diberi parameter t=time() agar tidak dicache browser -->
    

    <!-- Tombol refresh captcha -->
    <button type="button" id="refresh-captcha" style="cursor:pointer;"
      class="p-1 rounded-md text-gray-600 hover:bg-gray-100 hover:text-gray-900">
      
    </button>
  </div>

  <div class="relative">
    <div class="group relative">
      <!-- Input captcha dari user -->
      <input
        type="text"
        id="captcha"
        name="captcha"
        class="w-full h-[72px] pl-[80px] pr-6 pt-6 pb-2 font-semibold text-gray-900
border-[1.5px] border-gray-300 rounded-[24px] focus:outline-none focus:border-blue-500
peer transition-all"
        placeholder=" "
        required
      />

      <!-- Label yang akan naik ketika input aktif -->
      <label for="captcha"
        class="absolute left-[80px] top-1/2 -translate-y-1/2 text-gray-500 font-medium
peer-focus:top-4 peer-focus:text-sm
peer-placeholder-shown:top-1/2
peer-[&:not(:placeholder-shown)]:top-4
peer-[&:not(:placeholder-shown)]:text-sm
transition-all">
        Captcha
      </label>

      <!-- Icon captcha -->
      

      <!-- Garis pemisah icon & input -->
```

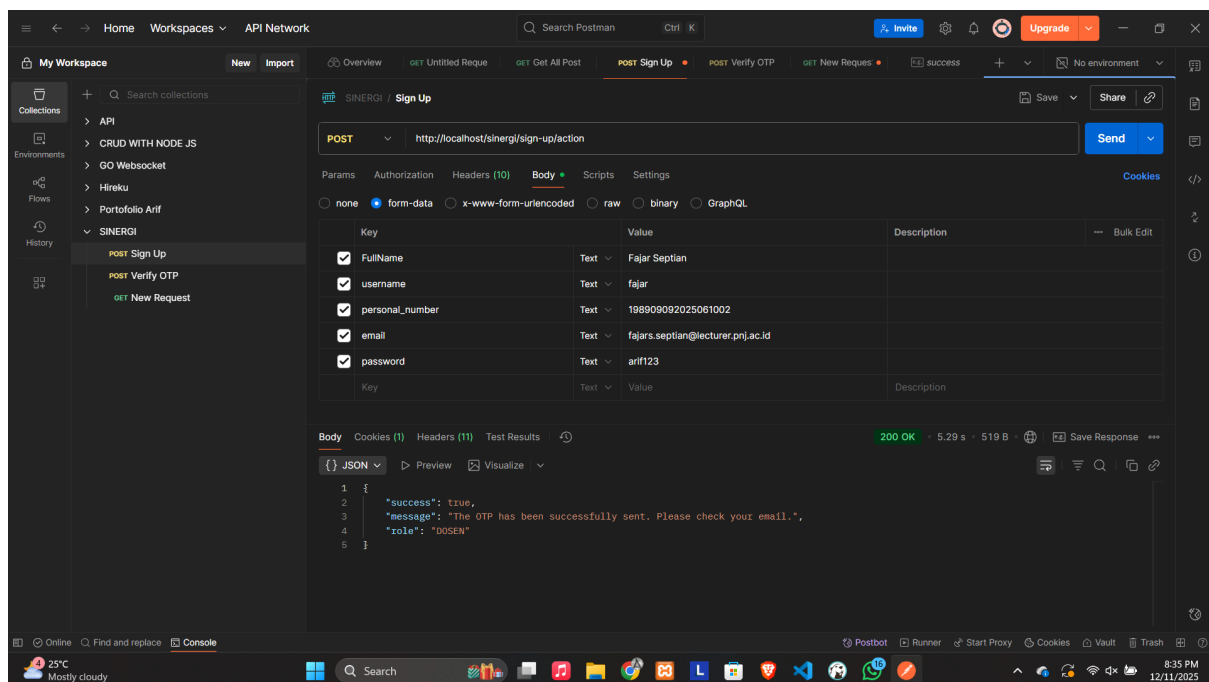
```

<div class="absolute left-[64px] top-1/2 -translate-y-1/2 w-[1.5px] h-6
bg-gray-300"></div>
</div>
</div>

<a href="<?php echo BASEURL; ?>/forget-password"
class="block text-end mt-3 font-medium text-gray-500 hover:text-blue-500
hover:underline">
    Forget My password
</a>
</div>

```

API



Endpoint <http://localhost/sinergi/sign-up/action>

Endpoint ini digunakan untuk melakukan pendaftaran akun baru (Sign Up).

API akan menerima data user, menentukan role secara otomatis (misalnya DOSEN), kemudian mengirimkan OTP ke email untuk proses verifikasi. Dengan Method POST.

Contoh Request (form-data)

FullName: "Fajar Septian"

username: "fajar"

personal_number: "198900902025061002"

email: "fajars.septian@lecturer.pnj.ac.id"

password: "arif123"

Success Response (200 OK) JSON Output

```
{  
  "success": true,  
  "message": "The OTP has been successfully sent. Please check your email.",  
  "role": "DOSEN"  
}
```